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सत्यमेव जयते

Department of Health Research
Ministry of Health and Family Welfare,
Government of India

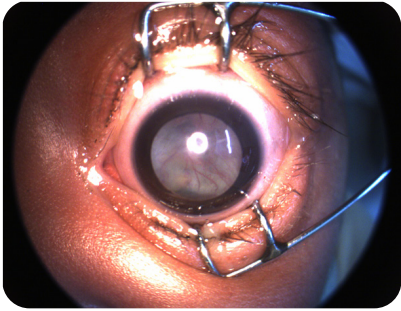


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Standard Treatment Workflow (STW)

RETINOPATHY OF PREMATURITY (ROP)

ICD-11-9B71.3



ROP is a potentially blinding retinal vascular disorder of preterm infants. Severe disease may cause retinal detachment and irreversible vision loss. Optimal quality neonatal care, timely screening and prompt treatment can prevent most ROP-related blindness.

WHOM TO SCREEN

Screen if any ONE of the following is present:

- Gestation <34 weeks
- Birth weight <2000 g
- Gestation 34-36 weeks with specified risk factors* or an unstable clinical course

***Significant cardiorespiratory support, prolonged/poorly controlled oxygen therapy, significant anemia, blood transfusion, sepsis, poor postnatal weight gain**

HOW TO ARRANGE SCREENING

- **Preferred:** On-site screening in SNCU/ NICU by an ROP-trained ophthalmologist/tele-screening team with appropriate monitoring
- **If unavailable:** Arrange scheduled specialist visits, or safe referral to the nearest designated ROP screening centre

TREATMENT INDICATIONS

- **Zone I:** Any stage with plus disease, or stage 3 without plus
- **Zone II:** Stage 2-3 with plus disease
- **Aggressive ROP (A-ROP):** Urgent treatment
- Reactivation/significant PAR after anti-VEGF; stage 2–3; stage 4–5
- **Advanced ROP (Stage 4-5):** Requires referral for surgery

Treatment-requiring ROP should be treated urgently, preferably within 48-72 hours of the decision to treat

WHEN TO SCREEN

First screen:

- By 4 weeks postnatal age
- By 2-3 weeks postnatal age, if GA <28 weeks or BW <1200 g
- If already overdue, screen ASAP
- If follow-up uncertain, screen before discharge even if not due

Repeat screen:

- As per advice by ROP-trained ophthalmologist according to retinal findings, usually every 1-3 weeks
- Posterior or progressive disease/suspected A-ROP may require review within 1 week or sooner

PREPARE TO SCREEN

- Withhold feeds - 1 hour before screening
- Dilate pupils: phenylephrine 2.5% + tropicamide 0.5-0.8% eye drops (2-3 times at 10 min interval)
- Swaddle, proparacaine eye drops, EBM/ oral 25% dextrose for analgesia
- Maintain temperature and monitor SpO₂

TREATMENT OPTIONS

- Laser photocoagulation and/or intravitreal anti-VEGF injection, as decided by the ROP specialist based on zone and disease characteristics.
- Vitreo-retinal surgery for advanced ROP

WHEN TO STOP SCREENING

- Stop only when advised by ROP-trained ophthalmologist (retina fully vascularised, or ROP fully regressed)
- After anti-VEGF treatment, prolonged follow-up at least until 65 weeks PMA is required because of the risk of late reactivation and persistent avascular retina (PAR)

HOW TO SCREEN

- Strict hand hygiene and asepsis
- Examine both eyes by indirect ophthalmoscopy (20D/28D lens); use speculum/scleral indentation as required
- Wide-field retinal imaging/tele-screening may be used where available
- Monitor for apnea, bradycardia, desaturation and feed intolerance
- Document ROP findings and the date/place of the next examination

PREVENTION

- Antenatal corticosteroids for fetal lung maturation
- **Safe oxygen therapy**
 - Use air-oxygen blenders and continuous pulse oximetry
 - Target SpO₂ : 91-95%
- Prevent and promptly treat sepsis
- Ensure mother’s own milk, optimal nutrition and adequate postnatal growth
- Avoid unnecessary blood transfusions

DOs

- Ensure timely ROP screening and follow-up
- Document the date and place of next screening in discharge card
- Counsel the family regarding need for follow-up and risk of vision loss if screening is delayed

DON'Ts

- DO NOT discharge/transfer an at-risk neonate without a documented ROP follow-up plan
- DO NOT delay referral/treatment when indicated
- DO NOT stop ROP follow-up without advice of ROP-trained ophthalmologist

KEY PERFORMANCE INDICATORS (KPIs)

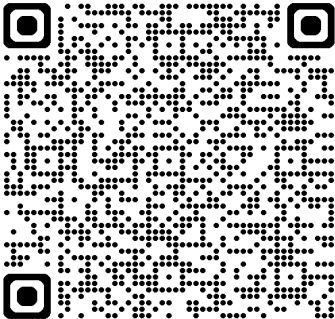
ROP screening coverage (%) = $\frac{\text{Eligible infants who received at least one ROP eye examination}}{\text{Total eligible infants for ROP screening}} \times 100$

Timely first ROP screening (%) = $\frac{\text{Eligible infants screened within the recommended first-screen window}}{\text{Eligible infants due for first screening}} \times 100$

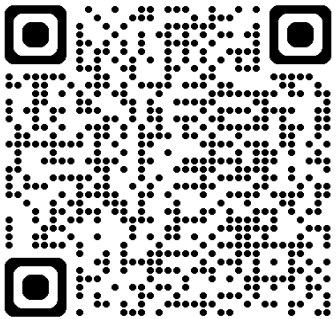
All SNCUs/NICUs should periodically assess the KPIs and undertake quality improvement measures to achieve 95-100% compliance

TIMELY ROP SCREENING SAVES SIGHT — NO ELIGIBLE INFANT SHOULD LEAVE THE SNCU/ NICU WITHOUT SCREENING OR A DOCUMENTED FOLLOW-UP PLAN

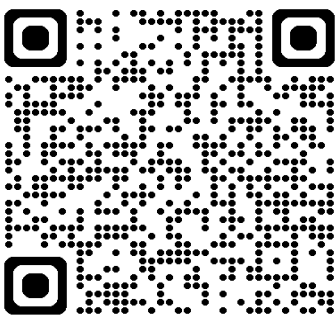
RELATED INFORMATION



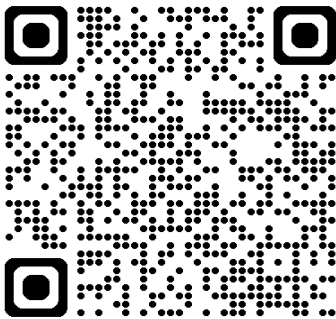
1. ROP CLASSIFICATION



2. ROP RECORD FORM



3. INFORMATION FOR PARENTS AND FAQs



4. PRETERM CARE PACKAGE

ABBREVIATIONS

A-ROP: Aggressive ROP
BW: Birth weight
EBM: Expressed Breastmilk

GA: Gestational Age
PAR: Persistent Avascular Retina
PMA: Postmenstrual Age

ROP: Retinopathy of Prematurity
VEGF: Vascular Endothelial Growth Factor

REFERENCES

1. MoHFW, Government of India. Guidelines for Universal Eye Screening in Newborns Including Retinopathy of Prematurity. RBSK. Available at: nhm.gov.in

2. National Neonatology Forum of India. Screening and Management of Retinopathy of Prematurity: Clinical Practice Guideline. Available at: nnfi.org/cpg.php

This STW has been prepared by national experts of India with feasibility considerations for various levels of healthcare system in the country. These broad guidelines are advisory, and are based on expert opinions and available scientific evidence. There may be variations in the management of an individual patient based on his/her specific condition, as decided by the treating physician. There will be no indemnity for direct or indirect consequences. Kindly visit the website of DHR for more information (**stw.icmr.org.in**) © Department of Health Research, Ministry of Health & Family Welfare, Government of India.